

# Chengyuan Luo

✉ [chengyul@andrew.cmu.edu](mailto:chengyul@andrew.cmu.edu)    [charles-cyluo.github.io](https://github.com/charles-cyluo)    [linkedin.com/in/chengyuan-luo](https://www.linkedin.com/in/chengyuan-luo)

## EDUCATION

---

### Carnegie Mellon University

Master of Science in Robotics

Pittsburgh, USA

Aug 2026 – Aug 2028 (Expected)

### Shanghai Jiao Tong University

Bachelor's degree in French, Associate degree in Information Engineering (minor)

Shanghai, China

Aug 2021 – Jun 2025

**GPA:** 4.12/4.3 (Minor), 3.98/4.3 (Overall)

**Selected coursework:** Machine Learning (98/100), Mathematical Foundation for Artificial Intelligence (98/100), Data Structure (98/100), Probability and Statistics (98/100), Digital Signal Processing (98/100)

### Honors and Awards:

- **Outstanding Graduate** of Shanghai Jiao Tong University, June 2025
- **Academic Scholarship (First Prize)** of SJTU-Paris Elite Institute of Technology (1/46), November 2024
- **Dean's Scholarship** of SJTU-Paris Elite Institute of Technology (1/73), September 2023
- **Gold Medal** in 46th International Collegiate Programming Contest (ICPC) Asia Regional Contest - Shanghai Site (rank 13/632) and Nanjing Site (rank 15/641), November and December 2021
- **Gold Medal** in 7th China Collegiate Programming Contest (CCPC) Weihai Site (rank 7/240), November 2021
- **Gold Medal** in China Computer Federation National Olympiad in Informatics Winter Camp (rank 18), August 2020

## PUBLICATIONS

---

### GBC: Generalized Behavior-Cloning Framework for Whole-Body Humanoid Imitation

*arXiv 2025*

Yifei He, **Chengyuan Luo**, Jiaheng Du, Wentao He, Jun-Guo Lu

### AnyPlace: Learning Generalized Object Placement for Robot Manipulation

*CoRL 2025*

Yuchi Zhao, Miroslav Bogdanovic, **Chengyuan Luo**, Steven Tohme, Kouros Darvish, Alán Aspuru-Guzik, Florian Shkurti, Animesh Garg

## RESEARCH EXPERIENCE

---

### AGIBOT Innovation (Shanghai) Technology, Embodied VLA Algorithm Intern

Mar 2026 – Jul 2026

#### Project 1: Synthetic Data Generation for VLA Policy Training

- ▷ Built a synthetic demonstration workflow for task-specific VLA policy training.
- Implemented simulation teleoperation to record expert demonstrations and validate task execution.
- Developed mimicgen-style trajectory generation with task-tailored trajectory randomization.
- Added lighting and background randomization to improve visual robustness and generalization.
- ▷ Expanded limited expert demonstrations into diverse multimodal VLA training data.

#### Project 2: Pico VR Teleoperation for Humanoid Robots

- ▷ Developed Pico VR teleoperation for intervention and task execution on AGIBOT's latest humanoid robot.
- Mapped controller buttons and axes to robot actions, including dexterous hand control and intervention.
- Tuned inverse kinematics parameters for task-specific manipulation and intervention behavior.
- Built a monitoring interface to track robot status and manage control and recording modules.
- ▷ Enabled intervention workflows for real-world dexterous manipulation on a specific task.

### ByteDance Seed Robotics, Intern

Oct 2025 – Feb 2026

#### Project: Large-Scale Data Generation for VLA Training

- ▷ Synthetic data generation for training VLA policies and evaluation of their performance in simulation.
- Developed general task setup enabling scalable configurations for new types of tasks.
- Optimized trajectory transform methods and improved inverse kinematics for tracking trajectories, increasing effective data generation efficiency by 48%.
- Improved time and memory efficiency of data generation simulation runs.
- ▷ Improved automated large-scale data generation pipelines that transformed a small number of human demonstrations into diverse, multimodal datasets optimized for VLA policies.

**SJTU Machine Vision and Autonomous System Laboratory**, Undergraduate Researcher *Dec 2024 – Jul 2025*  
Supervised by Prof. Jun-Guo Lu, Shanghai Jiao Tong University

**Project: Reinforcement Learning and Behavior Cloning for Bipedal Locomotion (GBC)**

- ▷ Integration of behavior cloning into humanoid robot control policies.
- Pre-processed motion capture datasets to extract reference actions with additional information.
- Developed an efficient buffer to store reference actions with NVIDIA Warp during reinforcement learning.
- Modified the protocols of the reinforcement learning library `rs1_rl` for behavior cloning reward functions.
- Conducted extensive training and hyperparameter tuning, leading to improved policy performance.
- ▷ The article has been published on ArXiv.

**People, AI, and Robotics (PAIR) Research Group**, Undergraduate Researcher (online) *Aug 2024 – Feb 2025*  
Advised by Prof. Animesh Garg, Georgia Institute of Technology.

**Project: Object Placement Simulation for AnyPlace**

- ▷ Implemented and simulated an object placement pipeline and evaluated the success rate.
- Modified the AnyGrasp model to generate grasps for diverse objects.
- Planned the pick-and-place trajectory using CuRobo to avoid collisions.
- Adapted the NVIDIA Isaac Lab grasping program for parallel executions of trajectories.
- Executed 20,000+ pick-and-place experiments across various objects and tasks.
- Analyzed the predicted placement poses and the simulated results to compute metrics for evaluation.
- ▷ The article has been accepted by **CoRL 2025**.

**SJTU Machine Vision and Intelligence Group**, Undergraduate Researcher *Feb 2023 – Dec 2024*  
Advised by Prof. Cewu Lu, Shanghai Jiao Tong University.

**Project: Benchmarking grasping models**

- ▷ Implemented an automatic framework to evaluate 2-finger grasp models using multiple metrics.
- Designed a novel framework for 2-finger grasp models to test their performance.
- Developed the grasp simulation based on the framework in Bullet and NVIDIA Isaac Lab environment.
- Implemented an entire pipeline for calibration and testing grasps in the real world using ROS and MoveIt Motion Planning Framework and conducted extensive experiments.
- Wrote a program to control a microcontroller unit using FreeRTOS for multithreading.
- ▷ The framework can execute grasps automatically with little human intervention, and it can evaluate grasps comprehensively using multiple metrics.

---

## WORK EXPERIENCE

**ABB Engineering (Shanghai) Robot Research Lab**, Research Intern for 3D Vision *Jun 2024 – Aug 2024*

**Project: Object detection and pose estimation**

- ▷ Identified objects and calculated their poses in a specific workspace.
- Implemented a framework to detect and estimate poses of specific objects using fiducial markers.
- Improved the detection using 2D object detection and segmentation models.
- Designed an algorithm to estimate poses using edge detection for objects with specific shapes.
- Complemented an additional academic survey on deep learning 3D reconstruction methods.
- ▷ Completed the internship's objectives with detailed documentation and several tests of the project.

---

## EXTRACURRICULAR ACTIVITIES

**SJTU RoboMaster Team**, Team Member *Oct 2022 – Aug 2023*

- Improved the detection of opponents' robots based on YOLO, aligned with the rule updates.
- Developed the detection for other contest apparatuses using YOLO and trained the neural network.
- Deployed the network on NVIDIA embedded AI computers and accelerated its efficiency using TensorRT.

---

## SKILLS

- **Programming Languages:** C/C++, Python, SQL
- **Software/Frameworks:** Linux (Arch Linux, Ubuntu), ROS, NVIDIA Isaac Sim and Lab,  $\LaTeX$
- **Languages:** Chinese, English (ETS GRE: 331 (V: 161, Q: 170) + 5.0), French (DELF B2)